



Technical Service Bulletin

Technical Service Bulletin: TSB210064

Released Date: 19-Jan-2026

T4 Offset Pin Piston

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK60 CM500
- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2350 K116
- QSK60 CM2350 K117
- QSK60 CM2350 K121
- QSK60 CM2350 K135
- QSK60 CM2350 K136

Description of Change

All QSK60 Advantage (CM2150 using CM2350 power cylinder) and CM2350 power cylinders will move at production and rebuild to offset pin steel pistons, and at repair events to new offset pin steel pistons.

Reason for Change

Offset pin steel pistons are available to improve the endurance of power cylinders within QSK60 CM2350 and Advantage engines in relation to power cylinder wall excitation.

Service Instructions

Piston Material Changes:

- FCD (Ferrous Cast Ductile) and steel pistons are **not** able to be mixed within an engine.
- FCD pistons can **only** be used with gas nitride cylinder liners.
- In the event of a malfunction requiring the installation of individual new pistons. An advantageous but **not** mandatory direction is to install steel offset pin pistons.
- Steel pistons are compatible with boron and gas nitride cylinder liners.

Piston Pin Offset Direction Change:

- Orientation of steel piston **must** be to the front of the engine according to the service manuals for any engine using offset steel pistons. An arrow is visible on the piston crown as a reference.
- From ESN#33241620 forward, factory CM2350 engines changed piston design to move the piston pin offset from the Thrust Side to the Anti Thrust Side. A new part number was released for the different offset.
- Mixing offset and non-offset pin steel pistons in individual engines is acceptable in fix as fail conditions. In engine rebuild conditions, the advantageous direction is to commonize all power cylinders to either offset and non-offset pin steel pistons throughout the individual engine (see last bullet point).
- All power cylinder components **must** be replaced on all power cylinders receiving offset steel piston kits.
- Use of steel offset pin pistons is advantageous in all engines passing through rebuild.
- The change to the offset pin piston will also drive a CPL change to CPL5849 for Advantage. As a result, the engine dataplate **must** be changed to reflect this. Verify the correct orientation of new pistons prior to installation.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Current Service Parts				
Piston – Anti Thrust side Offset Piston.	5540885	Yes	Yes	6613616
Piston Kit (Piston, Pin and Retaining Rings)	5693802	Yes	Yes	6668621
Piston (Rod and Piston Kit)	5542408	Yes	Yes	6662795
Historical Part Number Changes				
Piston Kit (Piston, Pin and Retaining Rings)	5579185	Yes	Yes	5693802
Piston Kit (Piston, Pin and Retaining Rings)	5405861	Yes	Yes	5693802
Piston Kit (Rod and Piston Kit)	5372212	Yes	Yes	5542408
Piston Kit - Marine - (Rod and Piston)	5592728	Yes	Yes	5542408

Table 1, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Piston (Individual Thrust Side Offset)	5371862	Yes	Yes	5540885
Piston (Individual Non Offset to Thrust Side Offset)	3649827	Yes	Yes	5371862

Part Compatibility

- In fix as fail conditions offset pin steel pistons and standard alignment pin steel pistons can be mixed on the same engine at any time.
- **Never** mix steel and FCD pistons on the same engine because of the weight differences between FDC and steel pistons. The different materials create an unbalancing of the rotating masses placed on the crankshaft.

Part Identification

Part numbers are stamped on all components for identification.

Part Inventory Action

Use the existing part number in applications that are **not** relevant to the application for which this TSB is being released.

Production Status

Implemented for production. See Table 2.

Table 2, Production Information		
ESN First	Build Date ¹	Plant
33241620	19 January 2026	Daventry
33224106 (Advantage)	2 March 2021	Daventry
33226172 (Tier 4)	20 September 2021	Daventry

Document History

Date	Details
2021-3-24	Module Created
2021-8-13	Updated title and added T4 designation.
2021-9-28	Updated Table 2 with 33226172 (Tier 4).

Date	Details
2021-11-17	Updated Description of Change, Service Instructions, and Part Compatibility sections.
2023-9-7	Updated Reason for Change, Service Instructions, and Part Compatibility.
2026-1-13	Updated Service Instructions and Service Parts.

Last Modified: 19-Jan-2026
